RETAIL ANALYTICS CASE STUDY

Where is the profit leaking?

Discount and profitability analysis of 9,994 orders from a US office-supply retailer

Vladimir · SQL (BigQuery) · Superstore dataset

THE QUESTION

Sales are growing, but profit isn't keeping up. Which discounts are costing us money?

9,994

order lines, 4 years of sales

$2.30M

total sales

12.5%

overall profit margin ($286K)

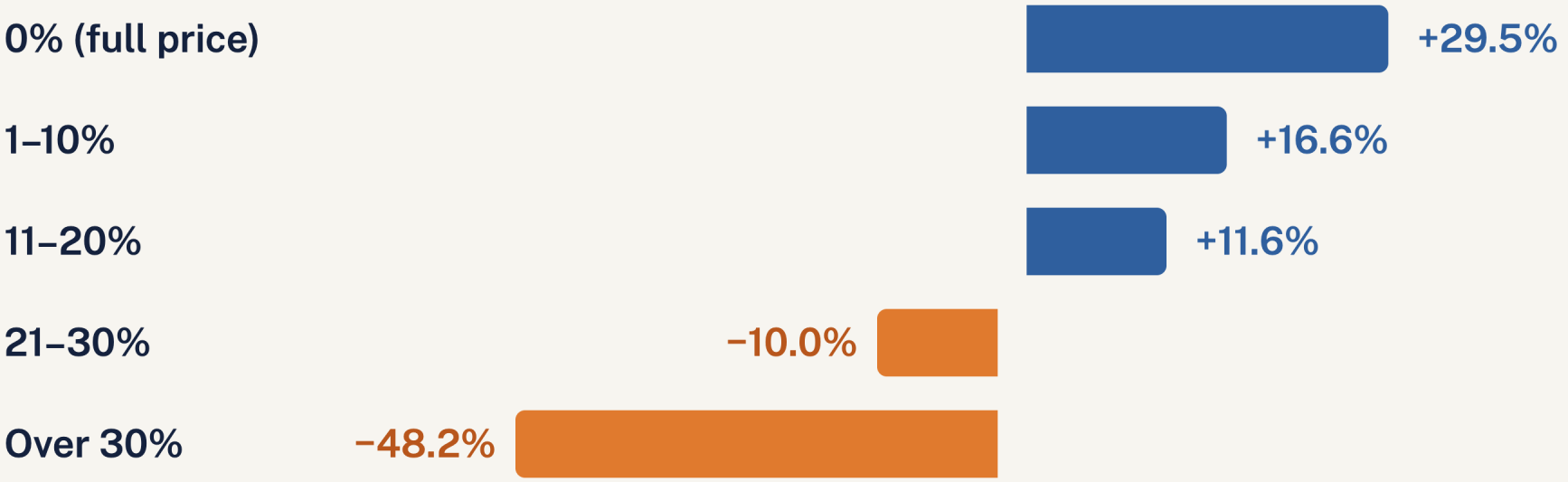
Data: Superstore Sales dataset (Kaggle). Orders grouped into five discount bands in BigQuery.

KEY FINDING

−$135K

lost on orders discounted more than 20%, equal to nearly half of total profit

Margin turns negative past 20% off



Profit margin by discount band. Blue bars make money; orange bars lose it.

Margin = profit ÷ sales per band. Source: Superstore dataset, BigQuery.

Full-price orders carry the business

Full price (0% off)

+$321K

47% of sales, but 112% of total profit. It earns more than the whole company keeps.

All discounted orders

−$35K

53% of sales, net loss. Over 30% off is just 12% of order lines but loses $125K.

Profit totals by discount band. Source: Superstore dataset, BigQuery.

Two recommendations

1

Cap discounts at 20%

Anything deeper needs manager approval. That's where margin flips negative, and those orders lost $135K.

2

Move promos to 11–20% off

Keep the deals customers want. This band still earns an 11.6% margin on $792K of sales.

Estimated impact

Conservative case

+$34K

Recover a quarter of the loss: about 12% more profit

Best case

+$135K

Deep-discount orders break even: about 47% more profit

Caveat: some customers won't buy at a smaller discount, so the real gain sits between these two. **Next step:** break losses down by sub-category to target the cap.

Current profit: $286K. Loss on orders over 20% off: $135K.